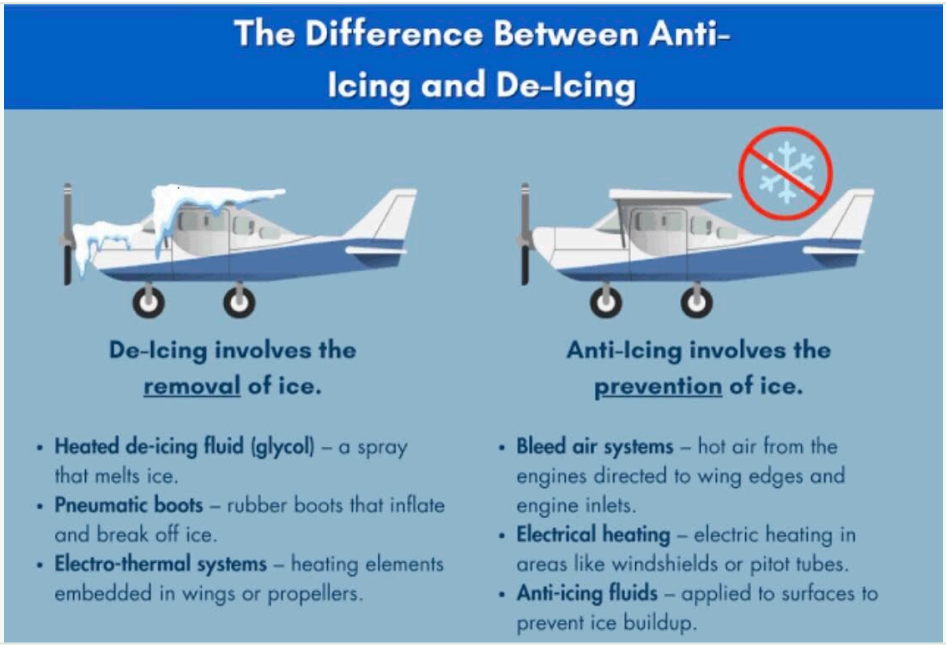
MODULE 2 · DE-ICING AND ANTI-ICING

Lesson 06: De-icing and Anti-icing Principles

Lesson 6 of 30 · about 8 min

LEARNING OBJECTIVES

- Distinguish de-icing from anti-icing.
- Describe one-step and two-step procedures and when holdover time starts.
- Identify the main safety considerations for treatment.



De-icing removes contamination; anti-icing prevents it for a limited time.

Two different jobs

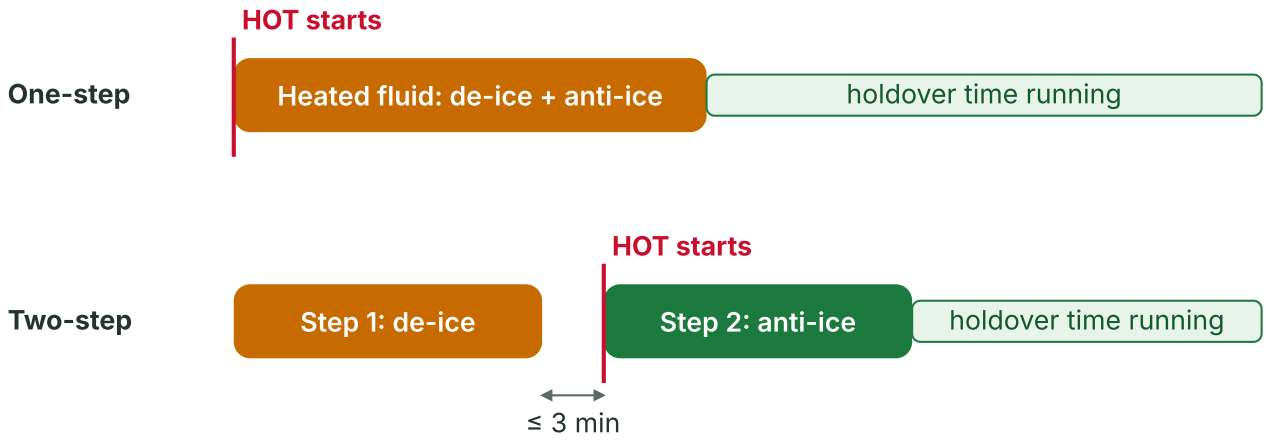
- **De-icing** removes frost, ice, snow or slush from the aircraft so that surfaces are clean. It is normally done with heated fluid (often Type I, or diluted Type II/IV), at a minimum of 60 °C at the nozzle. Hot water may be used at some temperatures if the procedure allows it. Some facilities use forced air or infrared heating.
- **Anti-icing** protects clean surfaces against new accumulation for a limited period — the **holdover time**. It is normally done with thickened fluid (Type II, III or IV), applied unheated or heated, onto a surface that is already clean.

When no precipitation is falling and no frost is forming, de-icing alone may be enough. When precipitation is falling, or is expected before take-off, anti-icing is required.

One-step and two-step procedures

One-step: a single application of heated, freezing-point-depressant (FPD) fluid cleans the aircraft and leaves a protective film. It is used when the expected holdover time from that fluid is enough.

Two-step: the first step removes the contamination; the second step applies an anti-icing fluid for longer protection. The second step must be applied **before the first-step fluid freezes** — typically within **3 minutes** — and treatment is done area by area if necessary.



The holdover time always starts at the **beginning of the final application**, not at the end.

Safety principles

- The aircraft must be configured as the AFM and de-icing procedures require before spraying starts (Lesson 9).
- No aircraft movement, engine power change or configuration change during treatment unless the de-icing crew is told and agrees.
- Treatment must be symmetrical: if one wing is treated, the other wing must receive the same treatment.
- Fluid makes the ramp extremely slippery for people and vehicles.
- Glycol fumes in the cabin or flight deck must be prevented by switching off bleed air as required; report any smell or smoke.
- A **post de-icing/anti-icing check** is mandatory after treatment (Lesson 12).

Responsibilities

The de-icing service provider is responsible for the quality of the treatment and the post-treatment check. The commander remains responsible for accepting the aircraft as clean and for deciding whether it remains clean until take-off.

KEY TAKEAWAYS

- De-icing removes contamination; anti-icing prevents new accumulation for a limited time.
- Two-step: second step within about 3 minutes, before the first-step fluid freezes.
- Holdover time starts at the beginning of the final application.
- Treatment must be symmetrical, and a post-treatment check is mandatory.

KNOWLEDGE CHECK

In a two-step procedure, when does the holdover time start?

A. At the start of step one

B. At the end of step one

C. At the beginning of step two (the final anti-icing application)

D. When the aircraft starts to taxi

Answer: C. Holdover time always starts at the beginning of the final (anti-icing) application.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.